Rahul Aditya Butta *Machine Learning Engineer*

butta.rahul.aditya@gmail.com +91 9505967830 Hyderabad, Telangana, India Portfolio

Professional Experience

2024/08 – 2024/11
Indore, India

Machine Learning Engineering Intern
Videoverse

- Developed a self-supervised learning pipeline for event detection in live cricket videos, leveraging video-masked encoder models to effectively extract and analyze temporal and spatial features.
- Integrated Anyscale's Ray into basketball event detection modules, enabling distributed computing to enhance system scalability, accelerate processing speed, and optimize resource efficiency for large-scale video data processing.
- Created a robust client and REST framework server to deploy and perform inference on a trained Visual Language Model (VLM), enabling efficient, scalable inference workflows.

2024/04 – 2024/07
Mumbai, India

Data Science Intern
Dronapay

- Developed a Retrieval-Augmented Generation (RAG) pipeline, fine-tuning an LLM to generate JSONLogic rules automatically from natural language inputs. Integrated FAISS for semantic retrieval of domain-specific variables, reducing manual rule-writing time.
- Fine tuned LLM's like Llama, Bert ect for different purposes like Named entity recognition, etc and deployed them into production using AWS lambda.
- Contributed on keystroke analysis based on machine learning, which strengthened system security and helped to reduce fraudulent actions by 12%.

2023/09 – 2023/11
Pune, India

Research Assistant
IIIT Pune

- Conducted extensive research in deep learning, specializing in object detection and image segmentation with help of Professor Dr. Sanjeev Sharma .
- Developed innovative algorithm to enhance the accuracy by 20% of an existing object detection system.
- Submitted our research findings on advanced techniques for water surface object detection at IEEE conference.

Education

2020 – 2024
Pune, India

Bachelor of Technology | Computer Science and Engineering
Indian Institute of Information Technology, Pune

- CGPA : 7.3

2018 – 2020
Mahbubnagar, India

Higher Secondary Education | Telangana Board of Intermediate education
Rishi Junior College

- Percentage: 95

Coding Profiles

Kaggle :
aditya_Rahul

LeetCode:
aditya_Rahul

Github:
buttaRahul

Skills

Programming Languages
Python, C++, SQL, PostgreSQL, Javascript, React.js, Django.

Machine Learning Tools
TensorFlow, PyTroch, Keras, scikit-learn.

Natural Language Processing(NLP) Tools
Spacy, Gensim, NLTK

Data Analysis
Pandas, Matplotlb, Seaborn

Projects

Enterprise Search
- An AI search solution for organizations, enabling unified search across web, databases, company sites, and emails. Built on RAG and LLM, it intelligently retrieves and generates relevant responses, streamlining information access
- Leveraged Langchain, Langchain-Tools, chromadb, Python, FastApi, React.js.
- Repository Link: https://github.com/buttaRahul/enterprise_serach

Query Assist
- Developed an NLP pipeline to covert Natural Language verbal specifiactions into sql queries and execute them against a database enabling seamless interaction with databases for users unfamilar with DBMS systems.
- Achieved a logical accuracy of 59% and an execution accuracy of 66% on WikiSQL dataset.
- Recognised with the best paper award for this project at INCITE 2024 conference on Adaptive Intelligence and subsequently published by Springer Nature as a chapter in Adaptive Intelligence.

Multimodal Video summarization
- Enhanced video summarization by integrating both textual and visual information. Instead of relying solely on the video transcript
- Leveraged Bert, langhcian, pyAV, opencv.
- Repository Link: https://github.com/buttaRahul/multimodal_video_summarization

Portfolio web page
- Personal Portfolio web page to view resume and all my projects.
- Implemented using React.js and Material UI.
- Demo Link: https://rahuladitya-portfolio.netlify.app/

Certificates

- Natural Language Processing by DataCamp
- Advance React.js by Meta
- Data Manipulation with pandas by DataCamp
- Frontend development by Meta

Accomplishments

- Won the best paper award for Query Assist project at INCITE 2024 by springer nature .
- Won 2nd prize in National Technical Paper Contest for Query Assist Project conducted by IETE
- Notebooks Expert in Kaggle and highest global rating of 920 in Kaggle Notebooks.
- 2566 Global Rank in Google Kickstart Round H .

Publications

QueryAssist: Multimodal Verbal Specifications to Structured Query Conversion Model Using Word Vector-Based Semantic Analysis
Springer Nature Link